



Research paper

A hybrid decision making support method for parcel lockers location selection

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ABSTRACT

Due to the rise of e-commerce, there has been a significant increase in commercial transactions, playing a crucial role in driving economic growth. However, the delivery systems need help locating addresses and dealing with issues such as missed deliveries and exacerbating problems in last-mile logistics, such as traffic congestion and environmental impact. Parcel lockers offer an efficient solution to address these critical issues, but selecting an appropriate location is vital to ensure optimal service and improve overall logistic performance. For this motive, our work aims to define the best location for parcel lockers in Ireland, Dublin, using the Spherical Fuzzy Analytic Hierarchy Process (SFAHP). Spherical Fuzzy Sets (SFS) involve decision-makers hesitating in evaluating location selection. The integrated model is a suitable approach to avoid AHP deterministic disadvantages. Moreover, it is the first application of SFAHP to select parcel locker locations. The obtained results will benefit policymakers in making their strategic decisions on selecting parcel locker locations.

1. Introduction

The expansion of e-commerce is largely to blame for the abrupt changes and developments in the last-mile delivery segment during the past two decades. Ireland's Business Customer (B2C) market is predicted to grow by 17.16% annually to reach US\$15,915.1 million in 2022. The medium to long-term increment story of the B2C Ecommerce industry in Ireland promises to be desirable. The B2C Ecommerce is predicted to grow steadily over the forecast period, recording a compound annual growth rate of 12.80% during 2022–2026. The country's B2C Ecommerce Gross Merchandise Value will increase from US\$13,583.9 million in 2021 to US\$25,769.1 million by 2026. Consumers want the freedom to buy whenever they want and from wherever they are; however, they are less likely than they once were to wait for products to arrive. Beyond this, Consumers want access to a service available around the clock, making delivery operations more challenging from the perspective of the courier, express, and parcel delivery companies (Forbes Communications Council, 2018).

Logistics Service Providers (LSP) may suffer from costs and idle time due to missed delivery. Home delivery has several inherent limitations. For example, since goods are delivered door to door rather than Business to Business (B2B), it needs to consolidate deliveries optimally. Home delivery rounds are also longer and have more loading and unloading

stops. The other drawbacks of urban freight movement, including congestion, pollution, and an increase in the likelihood of accidents, result from these restrictions (Browne et al., 2012).

Given that most individuals now work throughout the day and that home delivery may no longer be the optimal alternative, there is a need to create an option to please both the customer and the delivery companies (Emeç et al., 2016). In order to spot the light on these problems, postal operators have invested in value-added services such as delivery to monitored or automated collect points over the past ten years. In order to satisfy the expectations of final customers, these new delivery approaches provide a feasible substitute for home delivery. With the accelerated growth of the e-commerce market and the broadening of logistics services, several researchers spotted the importance of the parcel locker system as a significant part of the last-mile delivery in the supply chain system. As a self-service tool, the parcel locker system is considered an efficient system to solve the last mile case, which is a critical limitation of delivery, since parcel lockers enhance the delivery network system quality and increase customer satisfaction as well (Viu-Roig and Alvarez-Palau, 2020).

Yang et al. (2020) proposed the developed bilevel programming approach to evaluate the problem of parcel locker location, the proposed approach integrated with GIS data. Schaefer and Figliozzi (2021) employed cluster analysis to ameliorate the quality of Amazon lockers

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by selecting the optimal location. Ghaderi et al. (2022) conducted an integrated algorithm and a linear program approach to locate parcel lockers' locations in its simple version. Iwan et al. (2016) evaluated the quality of the parcel lockers system by measuring its efficiency and usability. Tang et al. (2021) performed a statistical analysis and confirmatory factor analysis to estimate the parcel lockers' service quality by defining the main influenced factors and their impact on consumer satisfaction. Pan et al. (2021) suggested an integrated Genetic Algorithm with the Lin-Kernighan Heuristic to estimate and enhance the efficiency of parcel lockers service quality. Keeling et al. (2021) evaluated the parcel lockers' location problem by adopting a multiple-criteria method.

Parcel locker location preferences vary depending on the type of location. Commercial areas were the best location for parcel lockers in Sweden (Bengtsson & Vikingson, 2015) and Brazil (Oliveira et al., 2019). In Poland, commercial areas and public transport stops were identified as the worst locations for parcel lockers (Iwan et al., 2016; Lemke et al., 2016), whereas, in Korea, public transport stops were the most preferred location (Lee et al., 2019). In France, the location preference differed between urban and rural areas. Public transport stops were the best location for parcel lockers in urban areas, while high population density areas were preferred in rural areas (Morganti et al., 2014). In Australia's South East Queensland, high population density areas with public transport stops were the most favoured location for parcel lockers (Lachapelle et al., 2018).

In this work, the problem has been structured in a hierarchical model and the popular Analytic Hierarchy Process (AHP) is employed with Spherical Fuzzy Sets (SFS) for the evaluation process to select the best location for parcel lockers. The AHP in a fuzzy environment generates significant and reliable results, and several works approved that in the case of location selection (Guler & Yomralioglu, 2020; Rajak & Shaw, 2019). The main advantage of using AHP with SFS is avoiding the hesitancy issue of an evaluator, where the hesitancy can be defined severally from membership and non-membership indexes (Kutlu Gundogdu & Kahraman, 2019). The conducted methodology has been successfully implemented to estimate parcel lockers' location selection in Dublin, Ireland. Furthermore, efficient insights have been presented for policymakers.

The remainder of this study is delivered as follows. Section 2 presents the performed AHP with SFS to assess the problem. An application for site selection of parcel lockers in Dublin is demonstrated in Section 3. In Section 4, the discussion and conclusion are illustrated.

2. Methodology

The AHP was proposed by Saaty (Saaty, 1977), and since that time, it has been improved and applied to solve complex problems in different fields (Dos Santos et al., 2019; Vargas, 1990) and specifically for transport-related complex problems (Moslem, Saraji, et al., 2023; Nassi & da Costa, 2012; Van Asch et al., 2020). AHP has been selected in our study since it is one of the most adopted multi-criteria decision-making (MCDM) approach for location evaluation problem (Dos Santos et al., 2019). For selecting the optimal location of parcel lockers, we proposed applying AHP in a Spherical Fuzzy environment; Kutlu Gundogdu and Kahraman (2019) introduced SFS to overcome the hesitancy issue of decision makers. The Spherical Fuzzy sets (SFS) are selected over linguistic or Pythagorean fuzzy sets because Spherical Fuzzy sets to let the evaluator generalise other extensions of fuzzy sets by defining a membership function on a spherical surface and independently assigning the parameters of that membership function with a larger domain (Kutlu Gundogdu & Kahraman, 2019). Shortly, Spherical Fuzzy sets synthesise linguistic or Pythagorean fuzzy sets by providing a general view of three-dimensional fuzzy sets. The summary of the main steps of the SFAHP model for the location selection evaluation process is depicted in Fig. 1.

The details of the evaluation process are presented step by step in the following seven main steps.

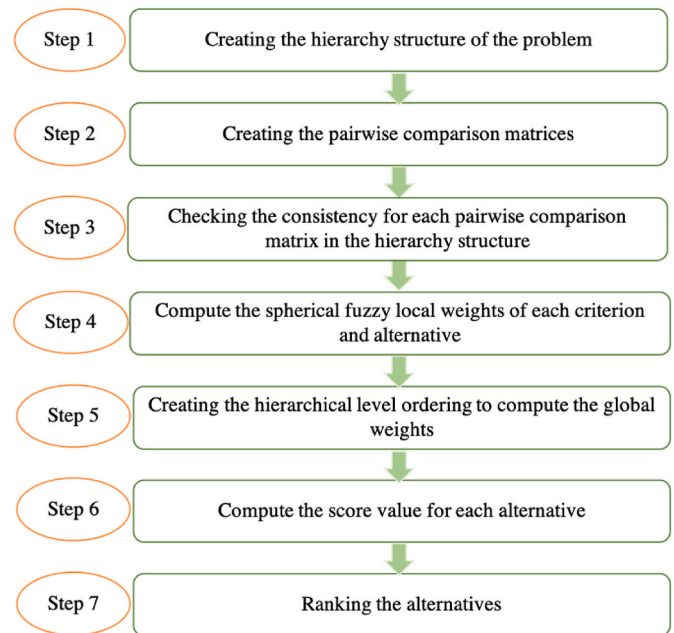


Fig. 1. The flowchart of the SFAHP method.

Step 1: Structuring the hierarchy model of the location selection problem.

In the first step, we create a hierarch structure based on specialists' perspectives and the literature. The hierarch structure generally consists of three levels, the first level is the target level (choosing the best location), the second level is the criteria level (the related criteria to the location selection problem, in some cases, it consists of several levels to represent the main criteria level and sub-criteria levels) and the third level is alternatives level (the suggested possible m alternatives $X = \{x_1, x_2, \dots, x_m\}$ ($m \geq 2$) for location selection). The criteria $C = \{C_1, C_2, \dots, C_n\}$ provides the weights and score index (SI).

Step 2: Construct the pairwise comparisons (PCs) employing the spherical fuzzy sets considering the linguistic terms (Table 1), where the evaluators used the crisp score indices or the linguistic scale to estimate the PCM. Equations (1) and (2) are employed to compute SI.

$$SI = \sqrt{100 * \left[(\mu_{A_i}^- - \pi_{A_i}^-)^2 - (v_{A_i}^- - \pi_{A_i}^-)^2 \right]} \quad (1)$$

for SMI; EI; AMI; VHI; and HI

$$\frac{1}{SI} = \sqrt{100 * \left[(\mu_{A_i}^- - \pi_{A_i}^-)^2 - (v_{A_i}^- - \pi_{A_i}^-)^2 \right]} \quad (2)$$

for VLI; ALI; EI; SLI and LI.

Table 1
The importance scale for PCs.

	(μ, ν, π)	Score Index (SI)
Absolutely more importance (AMI)	(0.9, 0.1, 0.0)	9
Very high importance (VHI)	(0.8, 0.2, 0.1)	7
High importance (HI)	(0.7, 0.3, 0.2)	5
Slightly more importance (SMI)	(0.6, 0.4, 0.3)	3
Equally importance (EI)	(0.5, 0.4, 0.4)	1
Slightly low importance (SLI)	(0.4, 0.6, 0.3)	1/3
Low importance (LI)	(0.3, 0.7, 0.2)	1/5
Very low importance (VLI)	(0.2, 0.8, 0.1)	1/7
Absolutely low importance (ALI)	(0.1, 0.9, 0.0)	1/9

Step 3: Detect the consistency for each PC matrix (PCM) to omit the inconsistent outcomes. To check the consistency, we replace the linguistic terms in the PCM with the corresponding SI. After that, we conduct the conventional process of computing the consistency ratio (CR). The CR threshold is 10% for PCM with (5 × 5) size matrices and smaller ones; however, it is 20% for PCM matrices bigger than (5 × 5).

Step 4: Computing the fuzzified local weights of each criterion and alternative.

The spherical fuzzy weights are calculated by using geometric mean (SWGMM), which is assigned as follows.

$$SWGMM_w(\bar{A}_{s1}, \dots, \bar{A}_{sn}) = \bar{A}_{s1}^{w_1} + \bar{A}_{s2}^{w_2} + \dots + \bar{A}_{sn}^{w_n} = \left\{ \prod_{i=1}^n \mu_{\bar{A}_{si}}^{w_i}, \left[1 - \prod_{i=1}^n (1 - \nu_{\bar{A}_{si}}^2)^{w_i} \right]^{\frac{1}{2}}, \left[\prod_{i=1}^n (1 - \nu_{\bar{A}_{si}}^2)^{w_i} - \prod_{i=1}^n (1 - \nu_{\bar{A}_{si}}^2 - \pi_{\bar{A}_{si}}^2)^{w_i} \right]^{\frac{1}{2}} \right\} \quad (3)$$

for w :

$$w = (w_1, w_2, \dots, w_n); w_i \in [0, 1]; \sum_{i=1}^n w_i = 1$$

Step 5: We are creating the hierarchical level ordering to compute the global weights of each criterion and alternative. The computation process is wisely applied from the lower to the top level.

The sore function is employed to adapt the crisp weights by using the following Eq. (4):

$$S(\bar{w}_j^s) = \sqrt{100 * \left[\left(3\mu_{\bar{A}_s} - \frac{\pi_{\bar{A}_s}}{2} \right)^2 - \left(\frac{\nu_{\bar{A}_s}}{2} - \pi_{\bar{A}_s} \right)^2 \right]} \quad (4)$$

The following Eq. (5) is applied to normalise the adopted weights:

$$\bar{w}_j^s = \frac{S(\bar{w}_j^s)}{\sum_{j=1}^n \bar{w}_j^s} \quad (5)$$

$$\bar{A}_{sij} = \bar{w}_j^s$$

$$\bullet \tilde{A}_{s_i} = \left\langle \left(1 - \left(1 - \mu_{\bar{A}_{s_i}}^2 \right)^{\bar{w}_j^s} \right)^{\frac{1}{2}}, \nu_{\bar{A}_{s_i}}^{\bar{w}_j^s}, \left(\left(1 - \mu_{\bar{A}_{s_i}}^2 \right)^{\bar{w}_j^s} - \left(1 - \mu_{\bar{A}_{s_i}}^2 - \pi_{\bar{A}_{s_i}}^2 \right)^{\bar{w}_j^s} \right)^{\frac{1}{2}} \right\rangle \forall i \quad (6)$$

Step 6: The final weight score of each alternative is fuzzily by using Eq. (4).

Step 7: For the defuzzified final weight scores, we rank the alternatives. The highest number refers to the best alternative.

3. Application to parcel lockers' location selection in dublin

The SFAHP has been adopted to evaluate the parcel lockers' location problem in Dublin City; the applied model is based on a hierarchy structure composed of criteria and alternatives. The justification for using Spherical Fuzzy sets over linguistic, or Pythagorean is not because Spherical Fuzzy set is the most recent extension of ordinary fuzzy sets but because Spherical Fuzzy sets let the evaluator generalise other extensions of fuzzy sets by defining a membership function on a spherical surface and independently assign the parameters of that membership function with a larger domain (Kutlu Gundogdu & Kahraman, 2019). Shortly, Spherical Fuzzy sets synthesise linguistic or Pythagorean fuzzy sets by providing a general view of three-dimensional fuzzy sets.

In consonance with a comprehensive literature review and scientists' points of view in the related field, the possibilities for locating parcel

lockers have been identified along with the related criteria, the specialists selecting the following six alternatives to be estimated as most preferred locations (A1: "Public transport stops"; A2: "Private car parking"; A3: "Post office"; A4: "On commercial area"; A5: "High population density area"; A6: "On pedestrian area"). The authors established the hierarchy structure of the parcel lockers' location selection (Fig. 2 illustrates the created structure of criteria and alternatives) based on the for mentioned alternatives and the following five main criteria.

C1: Reliability: indicates whether the quantity, size and providing quick service can meet consumers' daily demands. Also, a parcel lockers system is an efficient solution for non-delivery cases. Moreover, it can describe the dependability and consistency of a specific parcel locker location's ability to serve as a reliable pickup and drop-off point for users (Kilibarda et al., 2020).

C2: Accessibility: refers to the willingness to access the service point by walking, cycling, public transport or private vehicle, so the geographical location impact is important in positioning to make it easier for users to utilise. (Lagorio & Pinto, 2020).

C3: Traffic and operation: The parcel lockers system is a convenient solution for decreasing city traffic impact since the delivery will be oriented to lower points (Iwan et al., 2016).

C4: Security: refers to users' privacy and security while using the system, where the system can verify the delivery and simultaneously, the cameras will be recording what is happening 24/7, which is connected to police stations (Keeling et al., 2021).

C5: Environmental impact: Reducing delivery points by using parcel lockers could minimise the operated vehicles for the delivery process and decrease CO2 emission; many parcel lockers lead to lower environmental impact (Lagorio & Pinto, 2020).

In the evaluation process, three specialists estimated the PCMs (one 5 × 5 matrix for the criteria level and one 6 × 6 matrix for the alternative level) employing the linguistic terms from Table 1. The consistency for each PCM was revealed by considering the corresponding values in the conventional AHP approach for the linguistic terms from Table 1. Because of the complexity of the problem and the large size of the PCM, the current evaluation considered 0.2 as the CR threshold; the reason comes from the limited ability of the human to estimate the large size PCMs (Apostolou & Hassell, 1993).

The following tables (2–18) present the evaluated PCs, the detected CR, the computed spherical fuzzy weights $\bar{w}^s$, and the normalised crisp weights $\bar{w}^s$.

Appendix A. Shows the evaluated PCM of the criteria based on all decision-makers points of view concerning the goal (optimal location for parcel lockers). From the 1. decision maker (DM1) point of view, C4, followed by C2, were the most essential criteria.

From the 2. decision maker (DM2) point of view, C1 and C4 were the most essential criteria. However, from the 3. decision maker (DM3) perspective, C4, followed by C1, was the most essential criterion.

Appendix B. Shows the alternatives' estimated PCM based on all decision-makers points of view regarding the criteria. From the DM1 perspective and for the first criterion (C1), A1, followed by A3, were the most significant alternatives. The same priority was adopted from DM2

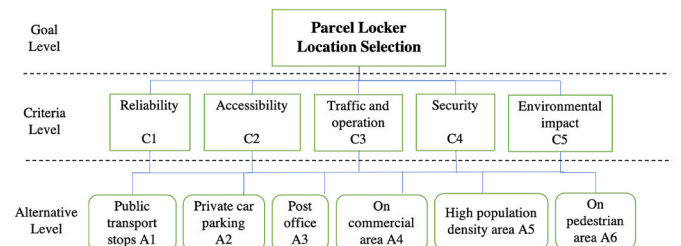


Fig. 2. The framework for choosing parcel locker location.

and DM3. Indeed, CR was acceptable for all evaluated PCMs.

From the DM1 perspective and for the second criterion (C2), A3, followed by A4, were the most significant alternatives. The same priority was adopted from DM2 and DM3. Indeed, CR was acceptable for all evaluated PCMs.

From the DM1 perspective and for the third criterion (C3), A1, followed by A2 and A3, were the most significant alternatives. The same priority was adopted from DM2; however, for DM3, A2, followed by A1 and A3, were the most significant alternatives. Indeed, CR was acceptable for all evaluated PCMs.

From the DM1 perspective and for the fourth criterion (C4), A4, followed by A3 and A2, were the most significant alternatives, and the same priority was adopted from DM3. From the DM2 perspective, A3, followed by A3 and A2, were the most significant alternatives.

From the DM1 perspective and for the fifth criterion (C5), A6, followed by A1 and A2, were the most significant alternatives; however, from the DM2 perspective, A6, followed by A2 and A3 were the most significant alternatives and from the DM3 perspective, A1 followed by A6 and A2 were the most significant alternatives.

Table 2 shows the weight score values for the alternatives based on the DM1 opinion, where A1 is the most suitable location, followed by A6 and A2.

Table 3 shows the weight score values for the alternatives based on the DM2 opinion, where A3 is the most suitable location, followed by A2 and A1.

Table 4 shows the weight score values for the alternatives based on the DM3 opinion, where A3 is the most suitable location, followed by A1 and A2.

Fig. 3 depicts the final aggregated weight score results of the alternatives computed using the geometric mean tool. The adopted results show that A3: "Post office" was the most suitable location for placing the parcel lockers with a weighted score of (0.196), followed by A1 "Public transport stops" with a weighted score of (0.178) and A2 "Private car parking" with a weighted score of (0.173), however, the worst option was A6 "On pedestrian area" with a weighted score of (0.129), followed by A5 "High population density area" with a weighted score of (0.156) and A4 "On commercial area" with a weighted score of (0.169).

3.1. Sensitivity analysis

For the validation and comparison of the proposed model, sensitivity analysis is a significant step to prove that the gained results of the evaluation are stable and implementable (Moslem, Stević, et al., 2023). The sensitivity analysis process is one of the AHP steps to apply and execute the AHP approach (Moslem et al., 2022; Çelikbilek & Moslem, 2023, pp. 1–26). This study's sensitivity analysis has been conducted as follows: In the criteria level, for the first DM point of view, the C4 criteria weight was increased from 0.243 to 0.250. The conducted modification in criterion weight from 0.243 to 0.250 tests the robustness and stability of the other criteria's weight, which is the target of applying sensitivity analysis. The weight scores of the other criteria in the criteria level were modified to maintain a score of one for the total calculation of weights.

Correspondingly, the C1 criteria weight was increased from 0.242 to 0.250 for the second DM point of view. The conducted modification in

Table 2

Composite impact of the parcel lockers location from DM1 point of view.

	0,209	0,224	0,129	0,243	0,196	Weight score
	C1	C2	C3	C4	C5	
A ₁	0,228	0160	0,201	0121	0,197	0202
A ₂	0,174	0138	0,207	0197	0,168	0171
A ₃	0,196	0213	0,177	0200	0,142	0149
A ₄	0,151	0200	0,185	0223	0,127	0122
A ₅	0,134	0190	0,151	0129	0,159	0162
A ₆	0,117	0100	0,080	0130	0,207	0194

Table 3

Composite impact of the parcel lockers' location from DM2 point of view.

	0,242	0178	0,134	0237	0,21	Weight score
	C1	C2	C3	C4	C5	
A ₁	0,233	0146	0,212	0119	0,193	0175
A ₂	0,18	0,141	0205	0,167	0204	0,176
A ₃	0,228	0223	0,166	0255	0,123	0205
A ₄	0,138	0208	0,176	0183	0,106	0163
A ₅	0,119	0185	0,132	0163	0,16	0,154
A ₆	0,102	0097	0,109	0114	0,214	0127

Table 4

Composite impact of the parcel lockers' location from DM3 point of view.

	0,256	0164	0,132	0265	0,182	Weight score
	C1	C2	C3	C4	C5	
A ₁	0,222	0167	0,219	0117	0,202	0180
A ₂	0,18	0,133	0222	0,158	0171	0,168
A ₃	0,213	0227	0,163	0205	0,149	0195
A ₄	0,153	0185	0,151	0197	0,122	0165
A ₅	0,137	0175	0,127	0188	0,162	0162
A ₆	0,095	0113	0,117	0134	0,194	0131

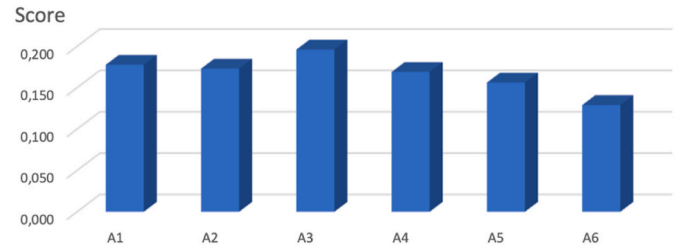


Fig. 3. The final aggregated scores for parcel locker's location alternatives.

criterion weight from 0.242 to 0.250 tests the robustness and stability of the other criteria's weight, which is the target of applying sensitivity analysis. The weight scores of the other criteria in the criteria level were modified to maintain a score of one for the total calculation of weights.

Correspondingly, the C4 criteria weight was increased from 0.265 to 0.270 for the second DM point of view. The conducted modification in criterion weight from 0.265 to 0.270 tests the robustness and stability of the other criteria's weight, which is the target of applying sensitivity analysis. The weight scores of the other criteria in the criteria level were modified to maintain a score of one for the total calculation of weights.

As depicted in Tables 5–7, the conducted change has not changed the criteria rank or order.

4. Discussion and conclusion

Place selection normally starts with defining the demand for usage and the demand for an additional capacity. The related decision is then made to demonstrate the analysis for the optimal location. Customer demand has integrated growing parcel volumes and demanded the perception of low shipping costs, and many retailers mask the true cost

Table 5

The criteria weights in the criteria level after sensitivity analysis for the DM1 point of view.

	$\bar{w}^s$	Rank	New weight after the Sensitivity Analysis	Rank
C1	0.209	3	0.207	3
C2	0.224	2	0.222	2
C3	0.129	5	0.127	5
C4	0.243	1	0.250	1
C5	0.196	4	0.194	4

Table 6

The criteria weights in the criteria level after sensitivity analysis for the DM2 point of view.

	$\bar{w}^s$	Rank	New weight after the Sensitivity Analysis	Rank
C1	0.242	1	0.250	1
C2	0.178	4	0.176	4
C3	0.134	5	0.132	5
C4	0.237	2	0.235	2
C5	0.210	3	0.208	3

Table 7

The criteria weights in the criteria level after sensitivity analysis for the DM3 point of view.

	$\bar{w}^s$	Rank	New weight after the Sensitivity Analysis	Rank
C1	0.256	2	0.255	2
C2	0.164	4	0.163	4
C3	0.132	5	0.131	5
C4	0.265	1	0.270	1
C5	0.182	3	0.181	3

of shipping to promote sales. This study proposed a new estimation system based on a decision support approach for the location selection of parcel lockers.

The parcel lockers' site selection problem must be systematically estimated under related elements. This requirement declaims employing a dynamic approach, such as the MCDM approach to estimate the related criteria and alternatives from specialists' points of view towards the problem. The most applied MCDM approach in the literature for site selection problems is the AHP approach; AHP-based measurement can provide quantity evaluators' subjective judgements. However, decision-makers usually need more support during the evaluation process; to accurately assess the impact of the elements, the AHP has been performed with spherical fuzzy sets.

In this study, five main criteria with six locations (A_1 : "Public transport stops"; A_2 : "Private car parking"; A_3 : "Post office"; A_4 : "On commercial area"; A_5 : "High population density area"; A_6 : "On pedestrian area") are used to select the optimal location of parcel lockers. The data are collected from three specialists in the transport and logistics field, each with at least five years of experience, the survey was delivered in October 2022, and the evaluators completed evaluating the survey in November 2022. The convenient SFAHP model successfully highlighted the post offices as the most suitable locations for placing parcel lockers; the spherical fuzzy extent efficiently determines criteria and alternative

priorities. The best location for parcel lockers in Dublin was the Post offices which are different from the best-experienced country in the world in parcel lockers usage, according to Iwan et al. (2016) and Lemke et al. (2016), where one of the best locations for parcel machines are shopping centres and malls areas.

In the future, we will focus on involving different stakeholder groups to deliver more sustainable solutions for parcel lockers' location selection problem. Moreover, extend the work by applying it to each district in Dublin City to get a real demand-based district. Indeed, adopted to save time and effort during the evaluation process, the Parsimonious AHP with Parsimonious Best Worst Method (Moslem, 2023) can be adopted.

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Authors' contributions

Sarbast Moslem: Conceptualization, Methodology, Analysis, Writing – review & editing, conceptualized the research and created the suitable methodology, conducted the survey, calculated the results. Francesco Pilla: Conceptualization, supervision and review. The author(s) read and approved the final manuscript,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Pairwise comparison of the criteria based on DM1 estimation.

	C1	C2	C3	C4	C5	$\bar{w}^s$	$\bar{w}^s$	$\bar{w}^s$	$\bar{w}^s$
C1	EI	EI	HI	SLI	EI	0,538	0410	0,345	0209
C2	EI	EI	VHI	SLI	EI	0,568	0395	0,322	0224
C3	LI	VLI	EI	LI	LI	0,339	0643	0,256	0129
C4	SMI	SMI	HI	EI	SMI	0,607	0378	0,301	0243
C5	EI	EI	HI	SLI	EI	0,507	0448	0,342	0196

CR = 0.045.

Pairwise comparison of the criteria based on DM2 evaluation.

	C1	C2	C3	C4	C5	$\bar{w}^s$			$\bar{w}^s$
C1	EI	EI	HI	SLI	EI	0,615	0378	0,282	0242
C2	EI	EI	VHI	SLI	EI	0,471	0510	0,325	0178
C3	LI	VLI	EI	LI	LI	0,359	0623	0,275	0134
C4	SMI	SMI	HI	EI	SMI	0,607	0378	0,301	0237
C5	EI	EI	HI	SLI	EI	0,544	0444	0,303	0210

CR = 0.087.

Pairwise comparison of the criteria based on DM3 evaluation.

	C1	C2	C3	C4	C5	$\bar{w}^s$			$\bar{w}^s$
C1	EI	EI	HI	SLI	EI	0,657	0337	0,242	0256
C2	EI	EI	VHI	SLI	EI	0,444	0543	0,300	0164
C3	LI	VLI	EI	LI	LI	0,366	0621	0,284	0132
C4	SMI	SMI	HI	EI	SMI	0,680	0310	0,243	0265
C5	EI	EI	HI	SLI	EI	0,489	0500	0,301	0182

CR = 0.091.

Appendix B

Pairwise comparison of locations caused by reliability based on DM1 evaluation.

C1	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	VHI	SMI	HI	HI	HI	0,683	0309	0,236	0228
A ₂	VLI	EI	SLI	HI	HI	HI	0,535	0467	0,269	0174
A ₃	SLI	SMI	EI	SMI	HI	HI	0,602	0389	0,285	0196
A ₄	LI	LI	SLI	EI	SMI	SMI	0,476	0516	0,301	0151
A ₅	LI	LI	LI	SLI	EI	SMI	0,424	0566	0,288	0134
A ₆	LI	LI	LI	SLI	SLI	EI	0,377	0606	0,286	0117

CR = 0.148.

Pairwise comparison of locations caused by reliability based on DM2 evaluation.

C1	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	HI	SMI	VHI	VHI	VHI	0,725	0270	0,205	0233
A ₂	LI	EI	SLI	HI	VHI	HI	0,577	0427	0,256	0180
A ₃	SLI	SMI	EI	HI	VHI	AMI	0,713	0288	0,219	0228
A ₄	VLI	LI	LI	EI	SMI	SMI	0,458	0541	0,284	0138
A ₅	VLI	VLI	VLI	SMI	EI	SMI	0,397	0605	0,268	0119
A ₆	VLI	LI	ALI	SLI	SLI	EI	0,349	0646	0,270	0102

CR = 0.111.

Pairwise comparison of locations caused by reliability based on DM3 evaluation.

C1	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	SMI	HI	VHI	VHI	0,691	0303	0,237	0222
A ₂	SLI	EI	LI	HI	HI	VHI	0,569	0438	0,244	0180
A ₃	SLI	HI	EI	HI	VHI	HI	0,665	0330	0,235	0213
A ₄	LI	LI	LI	EI	SMI	HI	0,495	0504	0,270	0153
A ₅	VLI	LI	VLI	SLI	EI	HI	0,443	0564	0,254	0137
A ₆	VLI	VLI	LI	LI	LI	EI	0,321	0667	0,239	0095

CR = 0.169.

Pairwise comparison of locations caused by reliability based on DM1 evaluation.

C2	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\widetilde{w}^s$			$\overline{w}^s$
A ₁	EI	SMI	LI	SLI	SLI	HI	0,515	0479	0,293	0160
A ₂	SLI	EI	VLI	VLI	LI	HI	0,443	0564	0,254	0138
A ₃	HI	VHI	EI	EI	SLI	VHI	0,662	0324	0,254	0213
A ₄	SMI	VHI	EI	EI	SMI	HI	0,631	0353	0,284	0200
A ₅	SMI	HI	HI	SLI	EI	HI	0,602	0389	0,285	0190
A ₆	LI	LI	LI	VLI	LI	EI	0,333	0652	0,248	0100

CR = 0.149.

Pairwise comparison of locations caused by reliability based on DM2 evaluation.

C2	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	LI	LI	SLI	SMI	0,476	0516	0,301	0146
A ₂	SLI	EI	VLI	LI	SLI	HI	0,455	0550	0,268	0141
A ₃	HI	VHI	EI	SMI	SMI	VHI	0,691	0303	0,237	0223
A ₄	HI	HI	SLI	EI	SMI	VHI	0,649	0346	0,252	0208
A ₅	SMI	SMI	SLI	SLI	EI	VHI	0,588	0408	0,285	0185
A ₆	SLI	LI	VLI	VLI	VLI	EI	0,327	0664	0,248	0097

CR = 0.094.

Pairwise comparison of locations caused by reliability based on DM3 evaluation.

C2	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	HI	SLI	SLI	SLI	SMI	0,525	0467	0,303	0167
A ₂	LI	EI	LI	SLI	SLI	SMI	0,424	0566	0,288	0133
A ₃	SMI	HI	EI	HI	HI	VHI	0,683	0309	0,236	0227
A ₄	SMI	SMI	LI	EI	SMI	HI	0,574	0419	0,293	0185
A ₅	SMI	SMI	LI	SLI	EI	HI	0,546	0448	0,293	0175
A ₆	SLI	SLI	VLI	LI	LI	EI	0,367	0619	0,279	0113

CR = 0.130.

Pairwise comparison of locations caused by reliability based on DM1 evaluation.

C3	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	SMI	SMI	HI	VHI	0,652	0340	0,269	0201
A ₂	SLI	EI	HI	HI	HI	VHI	0,665	0330	0,235	0207
A ₃	SLI	LI	EI	SLI	HI	VHI	0,577	0427	0,256	0177
A ₄	SLI	LI	SMI	EI	HI	VHI	0,602	0399	0,258	0185
A ₅	LI	LI	LI	LI	EI	VHI	0,495	0518	0,233	0151
A ₆	VLI	VLI	VLI	VLI	VLI	EI	0,281	0713	0,209	0080

CR = 0.191.

Pairwise comparison of locations caused by reliability based on DM2 evaluation.

C3	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	SMI	SMI	HI	VHI	0,652	0340	0,269	0212
A ₂	SLI	EI	SMI	SMI	HI	VHI	0,631	0363	0,269	0205
A ₃	SLI	SLI	EI	SMI	SMI	SMI	0,530	0458	0,320	0166
A ₄	SLI	SLI	SLI	EI	HI	HI	0,550	0445	0,285	0176
A ₅	LI	LI	SLI	LI	EI	SMI	0,424	0566	0,288	0132
A ₆	VLI	VLI	SLI	LI	SLI	EI	0,356	0633	0,272	0109

CR = 0.098.

Pairwise comparison of locations caused by reliability based on DM3 evaluation.

C3	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	HI	HI	HI	HI	0,660	0330	0,252	0219
A ₂	SLI	EI	HI	HI	HI	VHI	0,665	0330	0,235	0222
A ₃	LI	LI	EI	SMI	SMI	SMI	0,511	0482	0,301	0163
A ₄	LI	LI	SLI	EI	SMI	SMI	0,476	0516	0,301	0151
A ₅	LI	LI	SLI	SLI	EI	EI	0,411	0552	0,321	0127
A ₆	LI	VLI	SLI	SLI	EI	EI	0,382	0586	0,313	0117

CR = 0.091.

Pairwise comparison of locations caused by reliability based on DM1 evaluation.

C4	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	VLI	VLI	VLI	SLI	SMI	0,397	0605	0,268	0121
A ₂	VHI	EI	SLI	SLI	HI	SMI	0,628	0340	0,302	0197
A ₃	VHI	SMI	EI	SLI	HI	HI	0,628	0360	0,268	0200
A ₄	VHI	SMI	SMI	EI	HI	VHI	0,691	0303	0,237	0223
A ₅	SMI	LI	LI	LI	EI	SLI	0,424	0566	0,288	0129
A ₆	SLI	SLI	LI	VLI	SMI	EI	0,429	0564	0,294	0130

CR = 0.170.

Pairwise comparison of locations caused by reliability based on DM2 evaluation.

C4	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	LI	VLI	LI	SLI	EI	0,388	0579	0,305	0119
A ₂	HI	EI	SLI	EI	SLI	SMI	0,525	0467	0,303	0167
A ₃	VHI	SMI	EI	HI	VHI	AMI	0,757	0240	0,190	0255
A ₄	HI	EI	LI	EI	SMI	HI	0,569	0414	0,292	0183
A ₅	SMI	SMI	VLI	ALI	EI	SMI	0,514	0480	0,306	0163
A ₆	EI	SLI	ALI	LI	SLI	EI	0,376	0597	0,312	0114

CR = 0.099.

Pairwise comparison of locations caused by reliability based on DM3 evaluation.

C4	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SLI	LI	LI	LI	SLI	0,377	0606	0,286	0117
A ₂	SMI	EI	SLI	SLI	SLI	SMI	0,497	0490	0,321	0158
A ₃	HI	SMI	EI	SMI	SMI	HI	0,625	0363	0,285	0205
A ₄	HI	SMI	SLI	EI	SMI	HI	0,602	0389	0,285	0197
A ₅	HI	SMI	SLI	SLI	EI	HI	0,577	0416	0,285	0188
A ₆	SMI	SLI	LI	LI	LI	EI	0,424	0566	0,288	0134

CR = 0.089.

Pairwise comparison of locations caused by reliability based on DM1 evaluation.

C5	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	HI	HI	SMI	SMI	LI	0,595	0399	0,276	0197
A ₂	LI	EI	SMI	SMI	SMI	SLI	0,520	0470	0,311	0168
A ₃	LI	SLI	EI	SMI	SLI	SLI	0,449	0538	0,311	0142
A ₄	SLI	SLI	SLI	EI	SLI	LI	0,406	0575	0,311	0127
A ₅	SLI	SLI	SMI	SMI	EI	SLI	0,497	0490	0,321	0159
A ₆	HI	SMI	SMI	HI	SMI	EI	0,625	0363	0,285	0207

CR = 0.162.

Pairwise comparison of locations caused by reliability based on DM2 evaluation.

C5	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	SMI	SMI	VHI	SMI	SLI	0,613	0381	0,285	0193
A ₂	SLI	EI	VHI	VHI	SMI	SLI	0,637	0363	0,251	0204
A ₃	SLI	VLI	EI	SMI	LI	VLI	0,407	0592	0,275	0123
A ₄	VLI	VLI	SLI	EI	SLI	LI	0,356	0633	0,272	0106
A ₅	SLI	SLI	HI	SMI	EI	LI	0,515	0479	0,293	0160
A ₆	SMI	SMI	VHI	HI	HI	EI	0,668	0324	0,253	0214

CR = 0.124.

Pairwise comparison of locations caused by reliability based on DM3 evaluation.

C5	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆	$\bar{w}^s$			$\bar{w}^s$
A ₁	EI	HI	SMI	HI	SMI	SLI	0,602	0389	0,285	0202
A ₂	LI	EI	SMI	VHI	SMI	SLI	0,520	0470	0,311	0171
A ₃	SLI	SLI	EI	SMI	SLI	SLI	0,461	0524	0,322	0149
A ₄	LI	VLI	SLI	EI	SLI	SLI	0,382	0604	0,293	0122
A ₅	SLI	SLI	SMI	SMI	EI	SLI	0,497	0490	0,321	0162
A ₆	SMI	SMI	SMI	SMI	SMI	EI	0,586	0400	0,317	0194

CR = 0.151.

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